

Project Report: Blood Care Management System

Deployed On: <http://bloodcare.kesug.com>

1. Problem Description

The primary challenge addressed by this project is the lack of an integrated, real-time management system for blood bank operations. Manual record-keeping or fragmented systems often lead to:

- **Inefficient Inventory Tracking:** Difficulty in monitoring blood bag expiration dates and stock levels across different storage units.
- **Slow Response Times:** Delays in fulfilling urgent hospital requests due to poor communication and data visibility.
- **Donor Retention Issues:** Difficulty in tracking donor history to ensure safe intervals between donations.

Objective: To develop a three-tier web application that automates donor registration, supervises donations by staff, manages specialized storage units, and facilitates the transparent distribution of blood bags to hospitals via a transaction-based system.

2. Database Design

The database was designed using a relational approach to ensure data integrity and minimize redundancy.

Entity Relationship Diagram (ERD)

The system follows the **Chen Notation** for its ERD:

- **Entities:** Donor, Donation, Staff, Blood Bag, Storage Unit, Transaction, Hospital, and Request.
- **Key Relationships:**
 - **1:N (Donor to Donation):** One donor can make multiple donations over time.
 - **1:N (Staff to Donation):** One staff member supervises many donations.
 - **1:N (Donation to Blood Bag):** A single donation produces multiple blood components/bags.
 - **M:N (Blood Bag to Request):** Facilitated by the **Transaction** entity, where multiple bags can fulfill a single request.

Relational Schema (Table Structures)

The implementation includes several critical tables defined in `bloodcare_db.sql`:

- **Staff:** Stores administrative users with roles and secure passwords.
- **Donor:** Tracks details such as `BloodType` and `LastDonationDate`.
- **BloodBag:** Monitors the `Status` (Available, Quarantined, Expired) and `ExpirationDate`.
- **Hospital & Request:** Records formal needs from medical institutions and their fulfillment status.

3. Queries Used

The system utilizes dynamic SQL within the PHP backend to perform CRUD operations and generate analytical reports.

Data Manipulation (DML)

- **Insertion:** Used in `register_process.php` and `log_donation.php` to add new donors and donation events.
- SQL

```
INSERT INTO Donor (name, blood_type, phone) VALUES (?, ?, ?);
```

- **Updates:** Used to change blood bag statuses after distribution or expiration.
- SQL

```
UPDATE BloodBag SET status = 'Dispatched' WHERE bag_id = ?;
```

Data Retrieval & Joins

- **Joins:** Used in the various portals (Manager, Staff) to display comprehensive data. For example, linking donations to donors:
- SQL

```
SELECT Donor.name, Donation.donation_date, Donation.volume  
FROM Donation  
JOIN Donor ON Donation.donor_id = Donor.donor_id;
```

- **Aggregate Functions (Reporting):** Implemented in `backend/generate_report.php` to provide dashboard statistics:
- SQL

```
SELECT blood_type, COUNT(*) as TotalBags  
FROM BloodBag
```

```
WHERE status = 'Available'  
GROUP BY blood_type;
```

Advanced Database Objects

- **Indexes:** Created on frequently searched columns like `BloodType` to optimize query performance.
- **Views:** Virtual tables were implemented to allow managers to see "Available Stock" without viewing sensitive donor information.

4. Conclusion

The **Blood Care** project successfully implements a complete three-tier architecture. By integrating a normalized MySQL database with a responsive PHP backend and specialized portals for staff and hospitals, the system provides a robust solution for modern blood bank management.